
EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

OPERATION COLD IRON - Requirements & Grading Criteria

ARM Cortex-M33 • GDB • Ghidra

September 20, 2026

For educational and defensive security research only

OPERATION COLD IRON

REQUIREMENTS & GRADING CRITERIA

TARGET: NorthPharma Depot 7 cold store

ARTIFACT: ACT-I.bin / ACT-I.uf2 (compromised)

CREW: FROSTLINE OPERATIVE: NIGHTINGALE

Project Overview

NorthPharma runs the cold chain that keeps a vaccine lot viable. Each cold store is watched by a Pico 2 Cold Chain Monitor node. A sabotage crew called **FROSTLINE** planted an implant called **WHITEOUT** in the monitor image at Depot 7: a lying HMI, two open doors, and broken authenticated telemetry. Operative **NIGHTINGALE** recovered the compromised image as `ACT-I.bin`.

Students are the reverse-engineering reserve. They reverse engineer `ACT-I.bin` with Ghidra, find and patch all six defects, capture the runtime field key live in GDB, recover the provisioned passphrase and salt, export a corrected image, flash it to a real Pico 2, and prove the corrected behavior on the breadboard. The machine check is `scripts/verify_ctf.py`.

The challenge is a standalone capstone exercise and contains no answer, constant, address, bug, or patch belonging to any other course assignment.

Learning Objectives

- Decode an ARM Cortex-M33 vector and boot table and identify the reset handler and initial stack pointer.
- Map a stripped firmware image into modules by tracing calls from `main` and the monitor loop.
- Locate six corrupted bytes: a threshold immediate, an annunciator branch, a servo pulse constant, a remote-address gate, a tag-compare branch, and a nonce loop bound.
- Analyze Thumb immediate values, modified immediates, condition codes, and branch inversion.
- Capture a runtime-derived key with GDB and match it to an offline Argon2id derivation.
- Recover embedded provisioning secrets and state the OTP production requirement.

Students must use only the course concepts: ARM registers, stack behavior, USB-CDC and UART consoles, GDB, Ghidra static analysis and binary patching, vector tables, reset startup, XIP, Thumb addressing, data segments and literal pools, condition-code analysis, runtime key derivation, and the Argon2id plus XChaCha20-Poly1305 authenticated envelope.

Deliverables Checklist

#	Deliverable	Format	Criterion
1	Ghidra project screenshot	PNG/JPG	Task 1
2	Vector table and boot table	Inside <code>ACT-I-Answers.md</code>	Task 1

#	Deliverable	Format	Criterion
3	main and monitor-loop table	Inside ACT-I-Answers.md	Task 1
4	Module map	Inside ACT-I-Answers.md	Task 1
5	Cold offset evidence and patch	Inside ACT-I-Answers.md	Task 2
6	Dead annunciator evidence and patch	Inside ACT-I-Answers.md	Task 3
7	Damper geometry evidence and patch	Inside ACT-I-Answers.md	Task 4
8	Infrared door evidence and patch	Inside ACT-I-Answers.md	Task 5
9	Constant tag evidence and patch	Inside ACT-I-Answers.md	Task 6
10	Nonce reuse evidence and patch	Inside ACT-I-Answers.md	Task 7
11	GDB key capture	Inside ACT-I-Answers.md	Task 8
12	Provisioning recovery	Inside ACT-I-Answers.md	Task 9
13	ACT-I_fixed.bin	BIN file	Task 10
14	ACT-I_fixed.uf2	UF2 file	Task 10
15	Hardware proof and reflection	Inside ACT-I-Answers.md	Task 10

Required Tools and Equipment

Tool	Purpose
Raspberry Pi Pico 2	Isolated target node
Debug Probe (OpenOCD)	SWD connection for GDB inspection
arm-none-eabi-gdb	Runtime breakpoints and key capture
Ghidra	Static analysis and binary patching
Python 3 with uf2conv.py	UF2 conversion and artifact checks
DHT11, 1602 I2C LCD, RYL998, IR receiver, SG90 servo, 3 LEDs, button	Breadboard hardware proof
ACT-I.bin and ACT-I.uf2	Supplied compromised artifacts

Console settings: **USB-CDC virtual COM port, 115200 baud, 8 data bits, no parity, 1 stop bit**. Radio UART settings: **UART1, 115200, network ID 18**.

Artifact Identity

The instructor-issued artifact hashes are:

```

ACT-I.bin          b6fced1c0c305fdf167ed2da93464997504e1126d552eb120016d34c3a0bc47a
ACT-I.uf2          015e31dfeec6728afbc3a606d8b7b0283666f994acf124e56d0ed182119033f4
ACT-I_fixed.bin    e63c513a8b798c0e4be9265e4313b8d882f3b6fd8e9ae1161316adad7a926e29
ACT-I_fixed.uf2    f5df73b6ca69d4384ae64ffc39920e8d36634639a12b23904a962bd743332bf8

```

The verifier checks the ACT-I.bin and ACT-I.uf2 hashes specifically and asserts the six fixed bytes in ACT-I_fixed.bin.

Grading Rubric - Detailed Breakdown

Task 1: Setup and Initial Analysis (10 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Ghidra project created with the correct name and settings	2	Project ColdIron_Investigation , raw binary import	One item off	Not set up
[DOCUMENT] Processor configured as ARM Cortex 32 little endian default	2	Screenshot shows the correct processor	Wrong language	Missing
[DOCUMENT] Base address set to 0x10000000	2	Base 0x10000000	Wrong base	Missing
[DOCUMENT] Vector table, initial stack pointer, and reset handler identified	2	Base 0x10000000 , initial SP 0x20082000 , reset handler 0x1000015B	One missing	Not found
[DOCUMENT] main and the monitor loop (monitor_step) addresses identified	1	main 0x100001E0 , monitor_step 0x10006518	One correct	Neither
[DOCUMENT] Module map identifies the sensor, display, radio, status LEDs, button, servo, infrared, envelope, crypto, and KDF anchors	1	At least one correct anchor per module	Partial	Missing

Task 2: Bug #1 Cold Offset (12 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the cold offset compare at 0x1000759E	3	Address and function identified	Approximate	Not found
[DOCUMENT] Documented the compromised immediate and the correct threshold	3	Compromised 0xFA (250), correct 0x00 (0)	Correct address, no values	Wrong values
[DOCUMENT & PATCH] Patched 0xFA to 0x00 so a warm reading classifies as a breach	4	Byte 0xFA changed to 0x00	Wrong byte	Not patched
[DOCUMENT] Explained that a 5.0 C reading (50 tenths) now alarms	2	50 tenths at or above 0.0 C, so BREACH	Vague	Missing

Task 3: Bug #2 Dead Annunciator (12 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the annunciator branch at 0x1000754F	3	Address and function identified	Approximate	Not found
[DOCUMENT] Documented beq.n versus bne.n and the branch target	3	Correct branch and target 0x1000757E	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xD1 to 0xD0 so a breach routes to the red lamp	4	Byte 0xD1 changed to 0xD0	Wrong byte	Not patched
[DOCUMENT] Explained why the compromised branch leaves the HMI looking healthy	2	Breach falls into the green nominal path	Vague	Missing

Task 4: Bug #3 Damper Geometry (12 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the damper seal pulse constant at 0x10007686	3	Address and function identified	Approximate	Not found
[DOCUMENT] Documented the 800 versus 500 microsecond pulse and its angle	3	Compromised 800 us near 27 degrees, correct 500 us	Partial	Wrong
[DOCUMENT & PATCH] Patched 0x48 to 0xFA so the damper seals	4	Byte 0x48 changed to 0xFA	Wrong byte	Not patched
[DOCUMENT] Explained why the geometry error is a physical failure	2	The damper never fully closes so the store warms	Vague	Missing

Task 5: Bug #4 Open Infrared Door (12 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the infrared address gate at 0x10006569	3	Address and inlined handler identified	Approximate	Not found
[DOCUMENT] Documented the authorized address and the inverted branch	3	Address 0x1D (29), compromised beq.n	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xD0 to 0xD1 so only the maintenance address is accepted	4	Byte 0xD0 changed to 0xD1	Wrong byte	Not patched
[DOCUMENT] Explained why the compromised gate accepts every address	2	The reject branch is taken only on a match	Vague	Missing

Task 6: Bug #5 Constant Tag (12 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the tag-difference branch at 0x10007AFF	3	Address and function identified	Approximate	Not found
[DOCUMENT] Documented the constant-time compare and the branch condition	3	Zero difference means accept	Partial	Wrong
[DOCUMENT & PATCH] Patched 0xD1 to 0xD0 so forged frames fail authentication	4	Byte 0xD1 changed to 0xD0	Wrong byte	Not patched
[DOCUMENT] Explained why a non-zero tag difference must be rejected	2	A non-zero difference means forged or corrupt	Vague	Missing

Task 7: Bug #6 Nonce Reuse (12 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Located the nonce fill loop bound at 0x10007858	3	Address and function identified	Approximate	Not found
[DOCUMENT] Documented the compromised and correct loop bounds	3	Compromised 0x00, correct 0x18 (24)	Partial	Wrong
[DOCUMENT & PATCH] Patched 0x00 to 0x18 so 24 random bytes are drawn	4	Byte 0x00 changed to 0x18	Wrong byte	Not patched
[DOCUMENT] Explained nonce reuse and how a fresh random nonce stops it	2	Zero iterations means a fixed nonce; 24 random bytes fixes it	Vague	Missing

Task 8: Runtime Key Capture (8 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Breakpoint set at 0x100064CE after the crypto_kdf_argon2id call	3	Correct breakpoint address and command sequence	Address off	Not found
[DOCUMENT] Captured the 32-byte key from SRAM 0x20013074	3	32-byte hex read from g_key	Partial bytes	Missing
[DOCUMENT] Compared the captured key to the gateway Argon2id derivation	2	Captured key equals the derived key	Partial	Missing

Task 9: Provisioning Recovery (5 points)

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Recovered the passphrase and salt from flash	2	operation cold iron field key v1 and coldiron-salt-01	One recovered	Not found

Criterion	Points	Full credit	Partial credit	No credit
[DOCUMENT] Explained that committed secrets are dev-only and production must use OTP	1	OTP provisioning requirement stated	Vague	Missing
[DOCUMENT] Re-derived the key offline with Argon2id and matched the Task 8 capture	2	t=3, p=1, m=64 and a match	Partial	Missing

Task 10: Export and Verify (5 points)

Criterion	Points	Full credit	Partial credit	No credit
[PATCH] Exported ACT-I_fixed.bin from Ghidra	1	Valid patched binary	Corrupt	Not submitted
[PATCH] Converted to ACT-I_fixed.uf2 with the correct base and family	1	--base 0x10000000 --family 0xe48bff59	Wrong flags	Not submitted
[DOCUMENT] scripts/verify_ctf.py passes and hardware proves the red lamp, damper seal, and authentication	2	Verifier passes and the hardware proof is shown	Partial proof	No proof
[DOCUMENT] Reflection maps each of the six defects to a real-world cold-chain failure	1	Specific mapping for all six	Partial	Missing

Common Pitfalls

Pitfall	Consequence	Avoidance
Editing the wrong byte at 0x759E	Warm reading still nominal	The low byte is 0xFA ; change it to 0x00
Confusing beq and bne at 0x754F	Red lamp still dark	Fixed is beq.n (0xD0) to the red path
Treating the damper pulse as degrees	Wrong byte patched	The immediate is a pulse; 500 us is the seal
Missing that the IR handler is inlined	Cannot find the gate	Look inside monitor_step at 0x10006569
Reading the tag branch backwards	Forged frames still pass	Accept only on zero difference (beq.n , 0xD0)
Patching the nonce call instead of the bound	Loop still runs zero times	The bound is the immediate at 0x7858
Forgetting UF2 conversion	Raw binary will not flash	Use uf2conv.py with family 0xe48bff59
Fabricating the GDB key	Verification fails	The key equals the Argon2id output of Task 9

How To Breadboard

Device	Pin on device	Pico 2 GPIO	Notes
DHT11 data	DATA	GP4	10 kOhm pull-up to 3.3 V if the module needs it
1602 LCD	SDA	GP2	I2C1, backpack address <code>0x27</code>
1602 LCD	SCL	GP3	I2C1, 100 kHz
1602 LCD	VCC / GND	VBUS 5 V / GND	The backpack needs 5 V, not 3.3 V
RYLR998	RX	GP8 (Pico TX)	UART1, 115200, network ID 18
RYLR998	TX	GP9 (Pico RX)	UART1
IR receiver	OUT	GP5	VS1838B, internal pull-up enabled
Servo	signal	GP14	PWM 50 Hz; 1000 uF bulk cap across servo 5 V and GND
Red LED	anode	GP16	220 to 330 ohm to GND
Yellow LED	anode	GP17	220 to 330 ohm to GND
Green LED	anode	GP18	220 to 330 ohm to GND
Button	leg 1	GP15	Internal pull-up; leg 2 to GND, never to 3.3 V
Onboard LED	built in	GP25	Green blink on transmit
Debug Probe	SWCLK / SWDIO / GND	debug header	For GDB only

Use 3.3 V logic on every GPIO. The only 5 V connection is the LCD backpack supply. Keep the 1000 uF capacitor on the servo rail.

Memory Map Reference

Region	Address	Purpose
Bootrom	<code>0x00000000</code>	Immutable boot code
Flash/XIP	<code>0x10000000</code>	Vector table, code, rodata, data image
SRAM	<code>0x20000000</code>	Stack and writable state
Derived key <code>g_key</code>	<code>0x20013074</code>	32-byte Argon2id field key

The VA of any file offset is the file offset plus `0x10000000`.

Deadline & Submission

- Create a folder containing the Ghidra screenshot, `ACT-I_fixed.bin`, and `ACT-I_fixed.uf2`.
- Write all written answers in `ACT-I-Answers.md` inside that folder.

- Include the output of `python scripts/verify_ctf.py`.
- ZIP the folder as `lastname-firstname-ACT-I.zip`.
- Submit the ZIP before the posted deadline; late submissions lose 10 percent per day.

Grade Scale

Grade	Percentage	Points
A+	97-100%	97-100
A	93-96%	93-96
A-	90-92%	90-92
B+	87-89%	87-89
B	84-86%	84-86
B-	80-83%	80-83
C	70-79%	70-79
F	0-69%	0-69

Academic Integrity

Use only the supplied Pico 2 and firmware. Do not connect the exercise to an operational cold store, a pharmaceutical network, a public network, a military system, or a third-party device. This is a controlled, isolated educational exercise. All analysis and patches must be your own work; sharing binaries, addresses, keys, passphrases, or answers is a violation of the academic integrity policy.

Reference Material

Topic	Reference
ARM Cortex-M33 registers and stack	Course block 1
USB-CDC and UART console capture	Course block 2
Vector tables, reset startup, and XIP	Course block 3
Ghidra static analysis and binary patching	Course block 4
Condition-code and branch analysis	Course block 5
Servo geometry and actuator math	Course block 6
Argon2id and XChaCha20-Poly1305 authenticated envelope	Course block 7
Runtime key capture and provisioning	Course block 8